

JEE MAIN REVISION MASTER: ELLIPSE SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 116 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Ellipse - Core Application	116 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Ellipse - Core Application (Very High Yield — 116 PYQs)

Core Principle: Key problem pattern in Ellipse.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\frac{x^2}{a^2+7a+3}+\frac{y^2}{3a+15}=1$ represent an ellipse with major axis along y-axis, where f is a strictly dec..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a < b$, pass through the point $(4, 3)$ and have eccentricity $\frac{\sqrt{5}}{3}$..." [Ans: D]
- [2026 Evening Shift] Q#1 (MCQ): "If the points of intersection of the ellipses $x^2+2y^2-6x-12y+23=0$ and $4x^2+2y^2-20x-12y+35=0$ lie on a circle of radius r and ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

Let an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a < b$, pass through the point (4, 3) and have eccentricity $\frac{\sqrt{5}}{3}$. Then the length of its latus rectum is :

- (A) $\frac{4\sqrt{5}}{3}$
- (B) $2\sqrt{5}$
- (C) $\frac{7\sqrt{5}}{3}$
- (D) $\frac{8\sqrt{5}}{3}$ [Correct]

Correct Answer: D

Solution Analysis: Since $a < b$, the major axis is along the y-axis. So for the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, we use the eccentricity formula $e = \sqrt{1 - \frac{a^2}{b^2}}$. Given $e = \frac{\sqrt{5}}{3}$, so $e^2 = \frac{5}{9}$. Hence, $1 - \frac{a^2}{b^2} = \frac{5}{9}$ which gives $\frac{a^2}{b^2} = 1 - \frac{5}{9} = \frac{4}{9}$. Therefore, $a^2 = \frac{4}{9}b^2$. Now the ellipse passes through (4,3), so it satisfies the equation: $\frac{4^2}{a^2} + \frac{3^2}{b^2} = 1$. That is, $\frac{16a^2}{b^2} + \frac{9b^2}{b^2} = 1$. Substitute $a^2 = \frac{4}{9}b^2$: $\frac{16}{9} + \frac{9}{b^2} = 1$